

FINAL PROJECT REPORT

Digital Addiction&Screen Time Behavior Analysis

An interactive smartphone usage and digital well-being study using Microsoft Power BI

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Tool used	Microsoft Power BI Desktop	Date	September 2026
Dataset	smartphone_usage	Domain	Data Analytics / Digital Well-Being

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1. Introduction

1.1 Project Overview

Smartphones are now a part of everyday life. People use them to talk to friends, study, work, watch videos, play games and use social media. Because they are so useful, many people spend more time on their phones than they plan to. Too much screen time can lead to poor sleep, higher stress, lower focus at work or study, and unhealthy habits. This problem is often called **smartphone addiction** or **digital addiction**.

This project studies smartphone usage behavior using **Microsoft Power BI**. The dataset (smartphone_usage) has one record per user and includes age, gender, daily screen time, social media hours, gaming hours, work/study hours, sleep hours, notifications per day, app opens per day, weekend screen time, stress level, academic/work impact and addiction level. We cleaned this data, added a few helper columns, and built an interactive analytics solution on top of it.

The final Power BI file contains **five report pages**: a KPI page, a Visualization page, a Dashboard page, and two detailed report pages (Report1 and Report2). Together they use 51 visuals, including two AI-powered visuals (Decomposition Tree and Key Influencers) that help explain what factors are linked with addiction.

1.2 Objectives

The main goal is to turn raw smartphone usage data into clear and useful insights about digital addiction. The specific objectives are:

- To measure overall smartphone usage using KPIs such as average daily screen time, social media hours, gaming hours, app opens and notifications per day.
- To classify users by addiction level and to find how many users are addicted and how many are not.
- To compare screen time and usage habits across gender, age group and stress level.
- To study how heavy phone use is connected with sleep hours, stress level and work/study performance.
- To find the main factors that influence addiction using the Key Influencers and Decomposition Tree visuals.
- To design an easy-to-use dashboard and reports with slicers, and to give clear conclusions that support healthier phone habits.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Many people do not know how much time they really spend on their phones, or how it affects their sleep, stress and work. Data about usage exists, but in raw form (rows and columns) it is hard to read and does not answer questions directly. Decision makers such as parents, teachers, employers and well-being researchers need a simple visual way to see who is at risk, which habits matter most, and which groups need help.

The problem statement is described below from the point of view of three types of users.

No.	I am	I am trying to	But	Because	Which makes me feel
PS-1	A digital well-being analyst	Understand which users are addicted to smartphones	I only have raw data tables	The data has many columns and no visual summary	Unable to find patterns quickly
PS-2	A parent / teacher	See how phone use affects sleep, stress and study	I cannot see the connection between these factors	There is no simple comparison across age groups and stress levels	Worried and unsure how to help
PS-3	A decision maker (college / company)	Plan awareness programs for healthy phone use	I do not know which groups need the most attention	Addiction levels and usage patterns are not summarised	Uncertain about where to act first

Table 1: Problem statements

2.2 Project Proposal (Proposed Solution)

We propose an interactive Power BI solution that converts the `smartphone_usage` dataset into KPIs, charts, a dashboard and detailed reports. Users can filter the data by age group, gender, addiction level and stress level to see how usage patterns change.

Project overview	Analysis of smartphone usage and digital addiction using Power BI dashboards and reports.
Objective	Find who is addicted, what habits are linked with addiction, and how addiction affects sleep, stress and work/study.
Scope	User-level smartphone usage data. Analysis by age group, gender, stress level, addiction level and time spent on different activities.
Key features	KPI page with 11 KPI cards; Visualization page with 6 analysis charts; Dashboard with filter; 2 detailed report pages; Decomposition Tree and Key Influencers; slicers on all analysis pages.
Approach	Collect data, clean it in Power Query, add helper columns, build visuals, group them into a dashboard and reports, test filters and performance, then write conclusions.

Expected outcome	A clear, interactive view of digital addiction that shows risk groups and the factors behind addiction.
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Table 2: Proposed solution

Resource requirements

Resource	Details
Software	Microsoft Power BI Desktop (report design, Power Query, data model)
Data source	smartphone_usage dataset (structured table)
Hardware	Windows computer with at least 8 GB RAM
Optional	Power BI Service account to publish and share the report online

Table 3: Resource requirements

2.3 Initial Project Planning

The project was divided into clear phases. Each phase has its own activities and output.

Phase	Main activities	Output	Timeline
1. Initialization and planning	Understand the topic, define the problem statement, choose tools and prepare the plan	Problem statement, proposal, plan	Week 1
2. Data collection	Identify the dataset, load it into Power BI, review columns and data types	Raw data loaded in Power BI	Week 1
3. Data preprocessing	Check quality, fix data types, clean column names, create helper columns	Clean, analysis-ready table	Week 2
4. Data visualization	Frame business questions and build charts and KPI cards	KPI page and Visualization page	Week 3
5. Dashboard and reports	Arrange visuals, add slicers, add AI visuals, apply formatting and titles	Dashboard, Report1, Report2	Week 4
6. Testing	Test filters, cross-filtering, number of visuals and calculations, and page performance	Tested and stable report	Week 5
7. Documentation	Write insights, conclusion, future scope and final report	Final project report	Week 5

Table 4: Project plan (adjust the timeline to your actual schedule)

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The project uses one main data source, the **smartphone_usage** dataset. It is a structured table where each row describes one user record. The data was loaded into Power BI and appears in the model as the table named "smartphone smartphone_usage". Only one table is used, so the model has no relationships to manage.

Item	Details
Data source name	smartphone_usage dataset
Data type	Structured, tabular (rows and columns)
Loaded into	Microsoft Power BI Desktop (Get Data), table: smartphone smartphone_usage
Number of rows	7,500

Table 5: Raw data source

Fields used in the analysis

Field	Type	Meaning
user id	Identifier	Unique ID of the user
transaction id	Identifier	Unique ID of the record
age	Number	Age of the user
gender	Text	Gender of the user
daily screen time hours	Number	Average hours spent on the phone per day
social media hours	Number	Hours spent on social media per day
gaming hours	Number	Hours spent on mobile games per day
work study hours	Number	Hours of the phone used for work or study
sleep hours	Number	Hours of sleep per day
notifications per day	Number	Number of notifications received per day
app opens per day	Number	How many times apps are opened per day
weekend screen time	Number	Screen time hours on weekends
stress level	Category	Stress level reported by the user
academic work impact	Category	Effect of phone use on study or work
addiction level	Category	Level of smartphone addiction (used to group users)

Table 6: Original fields

3.2 Data Quality Report

Before building visuals, the data was checked for common quality problems. Good quality data is important because wrong or missing values can give wrong insights.

Quality check	What was checked	Action in Power Query / Power BI
Missing values	Blank or null cells in each column	Use Column Quality and Column Profile; remove or replace blanks
Duplicate records	Repeated rows or repeated IDs	Use Remove Duplicates on ID columns
Data types	Numbers stored as text, wrong types	Type detection was enabled; types checked and corrected
Outliers	Very high or very low hours (for example more than 24 hours a day)	Check min and max of hour columns
Category consistency	Different spellings of the same category	Review distinct values of gender, stress level and addiction level
Column names	Names with underscores and mixed style	Names made readable with spaces

Table 7: Data quality report

3.3 Data Exploration and Preprocessing

First, the data was explored using the Data view and Power Query Editor. We looked at column profiles, distinct values, minimum and maximum values and the distribution of each field. Then the following preprocessing steps were done:

- 1. Loaded the smartphone_usage dataset into Power BI.
- 2. Checked and fixed data types for every column (numbers, text and categories).
- 3. Made column names clear and readable (for example, "daily screen time hours").
- 4. Checked for missing values, duplicates and unusual values, and cleaned them where needed.
- 5. Created helper columns to make the analysis easier (see the table below).
- 6. Kept the model as a single table so that all filters and visuals work together without relationships.



Figure 4: Data preprocessing pipeline

Helper columns added to the table

Column	Purpose	Used in
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Column	Purpose	Used in
Age Group	Groups ages into ranges, so charts show a few clear groups instead of many single ages	Slicers, line charts, pie chart, column chart, Decomposition Tree, Key Influencers
Addicted Status	Marks each user as Addicted or not addicted	Dashboard bar chart, card filter (Addicted Status = Addicted), Key Influencers target
addicted label	Text label used to show Addicted vs Non-Addicted users	Donut chart on Visualization page
High Screen Time User	Flags users whose screen time is high	Donut chart on Report2
User Count	Number column used to count users easily	Donut chart on Report2

Table 8: Helper columns

4. Data Visualization

4.1 Framing Business Questions

Before creating charts, we wrote the business questions that the analysis should answer. Every visual in the project was built to answer at least one of these questions.

#	Business question	Visual(s) that answer it	Page
Q1	How many users are in the data and how many are addicted?	Total Users and Addicted Users cards	KPI'S, Dashboard
Q2	Do different genders use the phone for different lengths of time?	Average Daily Screen Time by Gender	VISUALIZATION, Dashboard
Q3	Which age group has the highest screen time?	Average Daily Screen Time by Age Group; weekend screen time by age group	VISUALIZATION, Report1
Q4	How are users spread across addiction levels?	Users by Addiction Level; pie chart of addiction level	VISUALIZATION, Dashboard
Q5	What share of users is addicted and what share is not?	Addicted vs Non-Addicted Users (donut)	VISUALIZATION
Q6	Does more screen time mean less sleep?	Daily Screen Time vs Sleep Hours (scatter); sleep by addiction level; sleep by age group	VISUALIZATION, Dashboard, Report2
Q7	Do gaming and social media have different effects?	Average Gaming vs Social Media Usage; social media hours by addiction level	VISUALIZATION, Report2
Q8	How is stress level connected with phone use?	App opens by stress level; stress level donut	Dashboard, Report2
Q9	How does phone use affect work or study?	Work/study hours by academic work impact	Report1
Q10	What behaviour is different on weekends?	Weekend screen time by addiction level and by addicted status	Report1, Dashboard
Q11	Which factors are most linked with addiction?	Key Influencers and Decomposition Tree	Report2, Report1

Table 9: Business questions mapped to visuals

The matrix below shows which pages answer each question. Every question is covered on at least one page.

4.2 Developing Visualizations

The visuals were developed on two pages: the **KPI'S page** for key numbers and the **VISUALIZATION page** for the main analysis charts.

A. KPI page

The KPI'S page has 11 cards. Each card shows one important number. Most cards show an average value, and two cards count users.

KPI card	Calculation	What it tells us
Total Users	Count of user id	Size of the dataset
Addicted Users	Count of user id, filtered to Addicted Status = Addicted	How many users are addicted
Average Age	Average of age	Typical age of users
Avg Daily Screen Time	Average of daily screen time hours	Overall phone usage per day
Avg Social Media Hours	Average of social media hours	Time spent on social media
Avg Gaming Hours	Average of gaming hours	Time spent on games
Avg Sleep Hours	Average of sleep hours	Overall sleep health
Avg Work/Study Hours	Average of work study hours	Productive use of the phone
Avg Notifications per Day	Average of notifications per day	Level of interruption
Avg App Opens per Day	Average of app opens per day	How often users check apps
Avg Weekend Screen Time	Average of weekend screen time	Weekend phone usage

Table 10: KPI cards

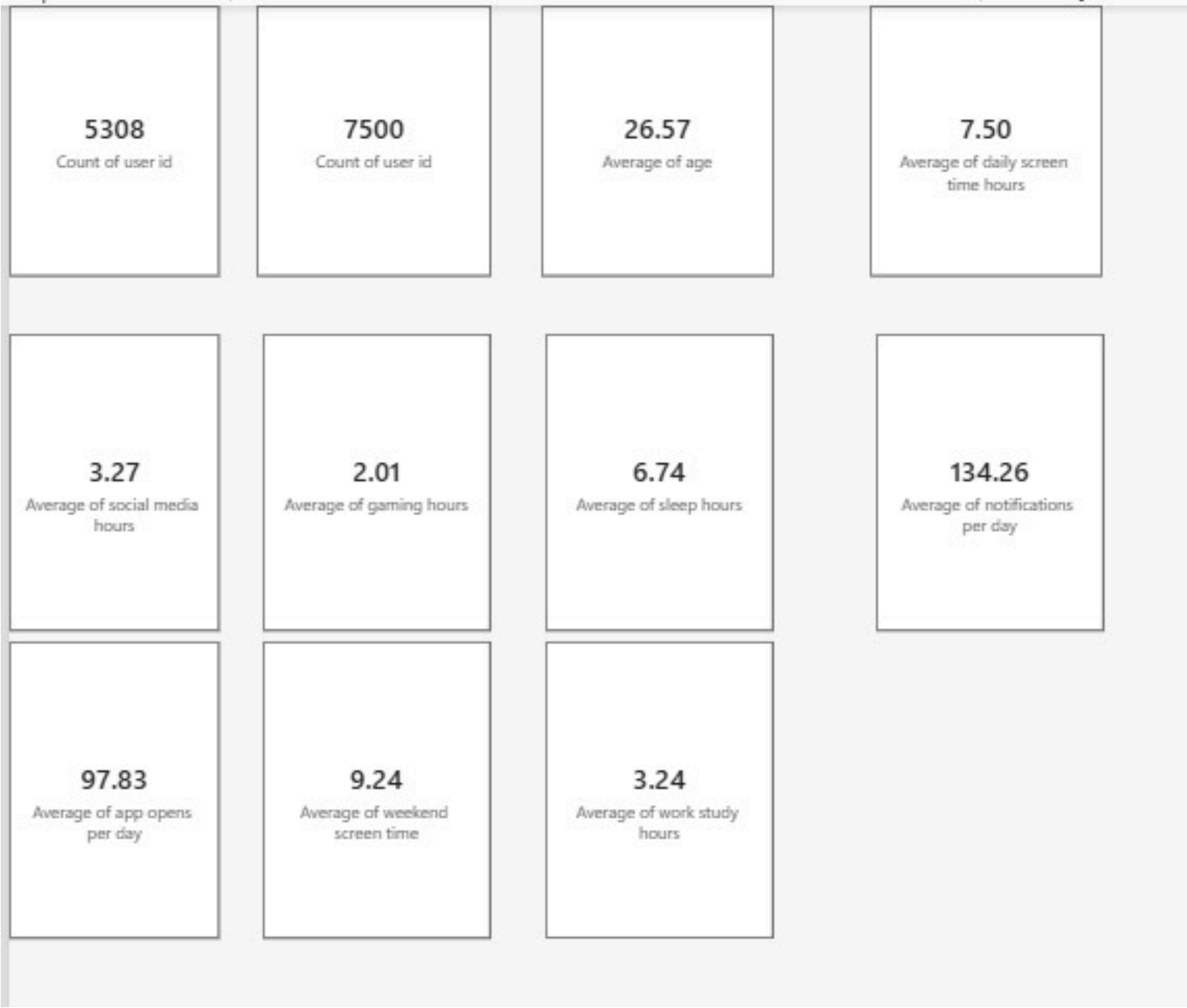


Figure 7: KPI page screenshot

Insights

- The dataset contains 7500 users, and 5800 of them are marked as addicted.
- The average user spends 7.5 hours per day on the phone, and 9.24 hours on weekends.
- Average sleep is 6.74 hours, while users open apps 97.83 times and receive 134.26 notifications per day.

B. Visualization page

The VISUALIZATION page has six analysis charts and three slicers (addiction level, age group and gender). All charts respond to the slicers.

Visual title	Type	Fields	Purpose
Average Daily Screen Time by Gender	Clustered column	gender, average of daily screen time hours	Compare screen time between genders
Users by Addiction Level	Clustered bar	addiction level, count of user id	Show how users are spread across addiction levels
Addicted vs Non-Addicted Users	Donut	addicted label, count of user id	Show the share of addicted users
Average Daily Screen Time by Age Group	Clustered column	Age Group, average of daily screen time hours	Find the age group with highest usage
Daily Screen Time vs Sleep Hours	Scatter	x: daily screen time hours, y: sleep hours	Check the link between screen time and sleep
Average Gaming vs Social Media Usage	Clustered column	average of gaming hours, average of social media hours	Compare how time is split between gaming and social media

Table 11: Visuals on the Visualization page

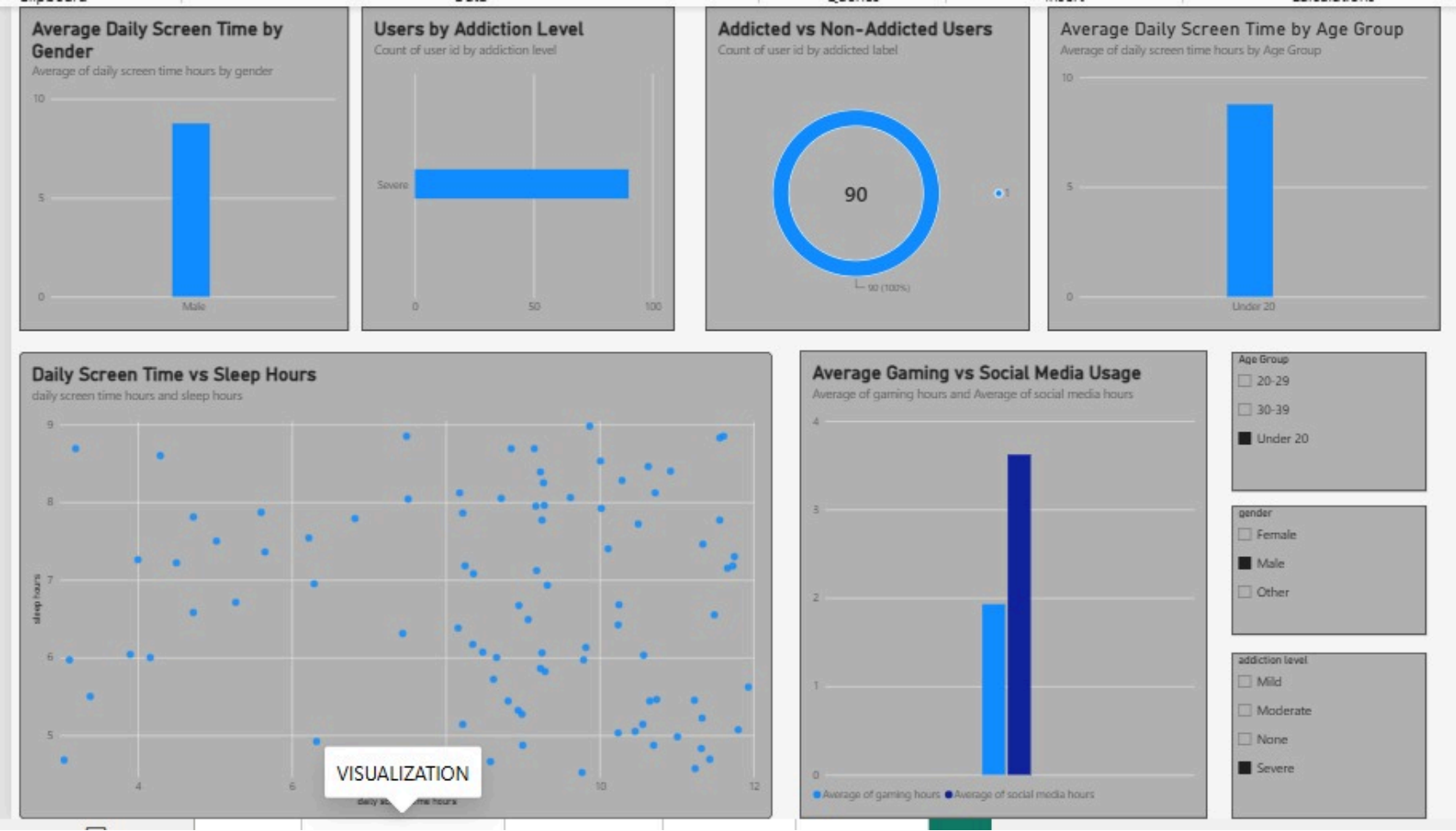


Figure 9: Visualization page screenshot

Finally, the chart below shows which fields are used most often across all visuals. Addiction level, user id and Age Group appear in the most visuals, so they are the main dimensions of the analysis. Helper columns (green) are used in many visuals, which shows why they were created.

5. Dashboard

5.1 Dashboard Design Screenshot with Insights

The **Digital Welbeing&Addiction Insight Dashboard** is the main summary page of the project. It gives a full picture of smartphone addiction on a single screen. The layout is simple: a title at the top, KPI cards below the title, and three rows of charts. One slicer for addiction level is placed in the top-right corner, so the user can filter the whole page.

Figure 11: Dashboard layout blueprint (generated from the report file)

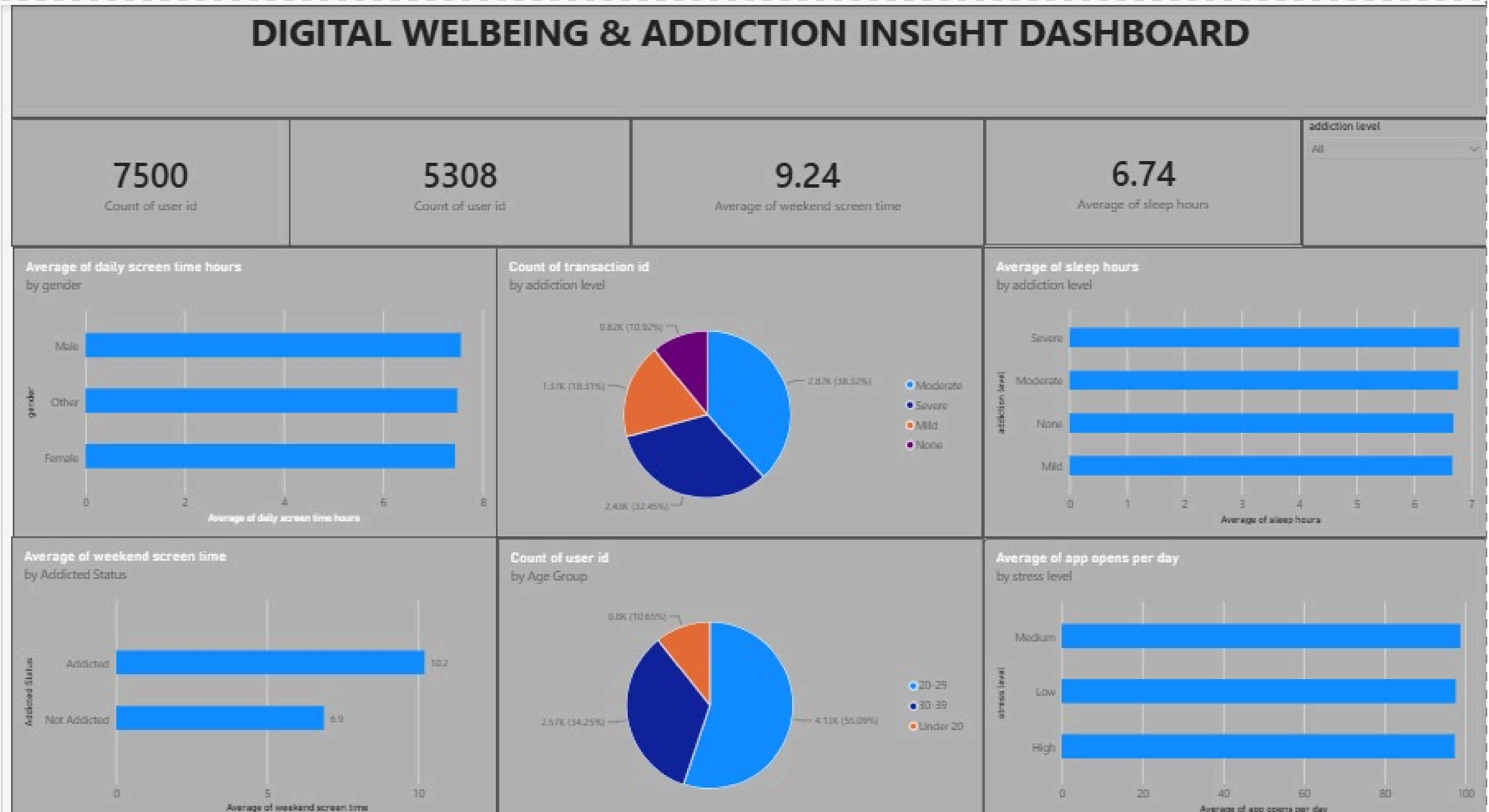


Figure 12: Digital Well-Being&Addiction Insight Dashboard (screenshot)

Dashboard components

Area	Visual	Fields	What it shows
Top row	4 KPI cards	Total users, Addicted users (filter: Addicted Status = Addicted), Avg weekend screen time, Avg sleep hours	Key numbers at a glance
Top right	Slicer	addiction level	Filters the entire dashboard
Row 2, left	Clustered bar	Avg daily screen time by gender	Gender comparison
Row 2, middle	Pie chart	Users by addiction level	Share of each addiction level
Row 2, right	Clustered bar	Avg sleep hours by addiction level	Sleep vs addiction

Area	Visual	Fields	What it shows
Row 3, left	Clustered bar	Avg weekend screen time by Addicted Status	Weekend habits of addicted vs other users
Row 3, middle	Pie chart	Users by Age Group	Age distribution of users
Row 3, right	Clustered bar	Avg app opens per day by stress level	Stress vs app checking behaviour

Table 12: Dashboard components

Interactivity: When a user selects an addiction level in the slicer, all cards and charts update at once. Clicking a bar or a pie slice also filters the other visuals (cross-filtering), so users can explore the data step by step.

6. Report

6.1 Report Design Screenshot with Insights

Two detailed report pages were created to go deeper than the dashboard. Report1 focuses on usage behaviour and includes a Decomposition Tree. Report2 focuses on well-being and includes the Key Influencers visual.

Report 1: Digital Addiction&Screen Time Behavior Analysis

This page starts with three KPI cards (average app opens per day, average social media hours and average notifications per day). Two slicers (age group and addiction level) sit next to them. Below are three bar charts, a line chart and a Decomposition Tree.

Figure 13: Report1 layout blueprint (generated from the report file)

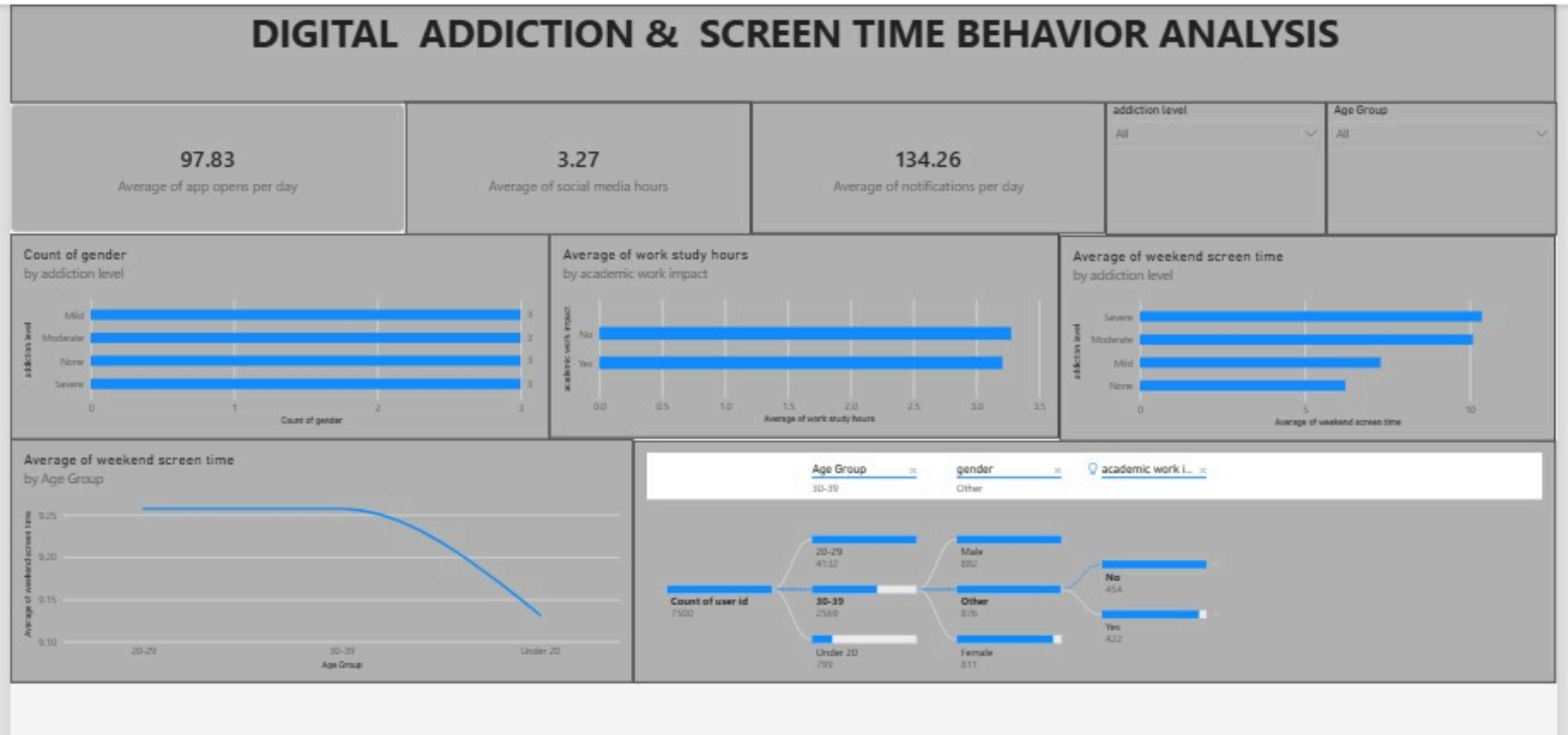


Figure 14: Report1 page screenshot

Visual	Type	Fields	Purpose
App opens, social media hours, notifications	3 KPI cards	Averages of app opens per day, social media hours, notifications per day	Show habit-related numbers
Slicers	2 slicers	Age Group, addiction level	Filter the whole page
Work/study hours by academic work impact	Clustered bar	academic work impact, average of work study hours	See how phone use effect relates to study/work time
Weekend screen time by addiction level	Clustered bar	addiction level, average of weekend screen time	Compare weekend usage across levels
Users by addiction level	Clustered bar	addiction level, count of records	See the distribution of users

Visual	Type	Fields	Purpose
Weekend screen time by age group	Line chart	Age Group, average of weekend screen time	See the trend across age groups
Decomposition Tree	AI visual	Analyze: count of user id. Explain by: Age Group, gender, academic work impact, stress level, addiction level	Break down users step by step to find important groups

Table 13: Report1 components

Report 2: Smartphone Addiction&Digital Well-Being Insights

This page focuses on well-being. It has KPI cards for average gaming hours, average age and average work/study hours, and slicers for gender and stress level. It also has a donut chart for stress level, a line chart for sleep by age group, a donut chart for high screen time users, a bar chart for social media by addiction level, and a **Key Influencers** visual.

Figure 15: Report2 layout blueprint (generated from the report file)



Figure 16: Report2 page screenshot

Visual	Type	Fields	Purpose
Gaming hours, average age, work/study hours	3 KPI cards	Averages of gaming hours, age, work study hours	Show user profile and usage
Slicers	2 text slicers	gender, stress level	Filter the whole page
Stress level share	Donut	stress level, count of user id	Show how users are spread across stress levels

Visual	Type	Fields	Purpose
Sleep hours by age group	Line chart	Age Group, average of sleep hours	Compare sleep across age groups
High screen time users	Donut	High Screen Time User, sum of User Count	Show share of high screen time users
Social media hours by addiction level	Clustered bar	addiction level, average of social media hours	Link social media use with addiction
Key Influencers	AI visual	Target: Addicted Status. Explain by: daily screen time hours, social media hours, gaming hours, weekend screen time, Age Group, notifications per day	Find which factors most increase the chance of addiction

Table 14: Report2 components

7. Performance Testing

Performance testing checks whether the report is fast, simple and easy to use. The report was tested for its filters, its calculations and the number of visuals on each page.

7.1 Utilization of Data Filters

Filters help users focus on the data they need and also keep the visuals light, because each visual only calculates the selected part of the data. The report uses slicers and visual-level filters.

Page	Filter / slicer	Field	Type
Dashboard	Slicer	addiction level	Slicer (whole page)
Dashboard	Filter on "Addicted Users" card	Addicted Status = Addicted	Visual-level filter
KPI'S	Filter on "Addicted Users" card	Addicted Status = Addicted	Visual-level filter
VISUALIZATION	Slicer	addiction level	Slicer
VISUALIZATION	Slicer	Age Group	Slicer
VISUALIZATION	Slicer	gender	Slicer
Report1	Slicer	Age Group	Slicer
Report1	Slicer	addiction level	Slicer
Report2	Text slicer	gender	Slicer
Report2	Text slicer	stress level	Slicer

Table 15: Filters used in the report

In total there are **8 slicers** and **2 visual-level filters**. Cross-filtering between visuals is also active, so a click on one chart updates the others.

7.2 Number of Calculated Columns / Measures

Item	Count	Details
Calculated / helper columns	5	Age Group, Addicted Status, addicted label, High Screen Time User, User Count
Custom DAX measures	0	All numbers use built-in aggregations (Average, Count, Sum) directly on columns
Relationships	0	Single-table model

Table 16: Calculations in the model

Using built-in aggregations and a single table keeps the model small and fast. The Power BI file is only about 0.3 MB, so it opens quickly and visuals refresh almost immediately when a slicer is changed.

7.3 Number of Visualizations

Page	Cards	Slicers	Charts	AI visuals	Total
KPI'S	11	0	0	0	11
VISUALIZATION	0	3	6 (3 column, 1 bar, 1 donut, 1 scatter)	0	9
Dashboard	4	1	6 (4 bar, 2 pie)	0	11
Report1	3	2	4 (3 bar, 1 line)	1 (Decomposition Tree)	10
Report2	3	2	4 (2 donut, 1 line, 1 bar)	1 (Key Influencers)	10
Total	21	8	20	2	51

Table 17: Visuals per page (text boxes used for titles are not counted)

8. Conclusion / Observation

This project shows how Power BI can turn a raw smartphone usage dataset into a clear, interactive story about digital addiction. Through five report pages, 51 visuals and simple filters, users can see how much time people spend on their phones, who is addicted, and how usage is linked with sleep, stress and work/study.

What we learned

- How to prepare data with Power Query and create helper columns.
- How to choose the right visual for each business question.
- How to use slicers, filters and cross-filtering to make reports interactive.
- How AI visuals such as Key Influencers and Decomposition Tree can find hidden patterns.

Recommendations: Users with high weekend and social media usage can be given awareness programs, screen-time limits and notification controls. Schools, colleges and workplaces can use such dashboards to spot risk groups early and support healthier sleep and study habits.

9. Future Scope

- Connect the report to live or regularly updated data so the dashboard refreshes automatically.
- Publish the report to Power BI Service and share it with teams, teachers or parents.
- Add DAX measures such as addiction rate %, average sleep gap and month-over-month change.
- Add time-based data (daily or weekly logs) to study trends and habits over time.
- Add more fields such as app category, location, income or occupation for deeper analysis.
- Build a machine learning model (for example in Python or Azure ML) to predict addiction risk.
- Add row-level security and mobile-friendly layouts for wider and safer use.
- Add automatic alerts when addiction rate or screen time crosses a safe limit.
- Include survey feedback and well-being scores to combine numbers with user opinions.

10. Appendix

10.1 Source Code

The project was built with the Power BI visual interface, so there is no separate program code. The important logic is in the helper columns. Please replace the examples below with your own formulas if they are different.

```
create database smartphone_usage;
```

```
use smartphone;
```

```
select * from smartphone_usage;
```

```
-- Gender level addiction counts
```

```
SELECT gender, COUNT(*), SUM(CASE WHEN addicted_label = 1 THEN 1 ELSE 0 END) AS addicted_count
FROM smartphone_usage;
```

```
GROUP BY gender;
```

-- Baseline usage averages

```
SELECT AVG(daily_screen_time_hours), AVG(sleep_hours), AVG(stress_level = 'High')
FROM smartphone_usage;
```

-- Null/mising-value check

```
SELECT * FROM smartphone_usage
WHERE age IS NULL OR daily_screen_time_hours IS NULL OR addiction_level IS NULL;
```

-- Stress level comparison by addiction status

```
SELECT addicted_label, stress_level, COUNT(*)
FROM smartphone_usage
GROUP BY addicted_label, stress_level
ORDER BY addicted_label, stress_level;
```

-- Addiction level distribution

```
SELECT addiction_level, COUNT(*), AVG(daily_screen_time_hours), AVG(sleep_hours)
FROM smartphone_usage
GROUP BY addiction_level;
```

-- Screen time comparison by academic/work impact

```
SELECT academic_work_impact, AVG(daily_screen_time_hours), AVG(social_media_hours),
AVG(gaming_hours)
FROM smartphone_usage
GROUP BY academic_work_impact;
```

Example DAX for helper columns (adjust the rules to match your model)

10.2 GitHub&Project Demo Link

Item	Link
GitHub repository	https://github.com/Suneetha121/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics.git
Project demo video	https://drive.google.com/file/d/1JsMI35yvago60sw9_L1fD3PEB66US02e/view?usp=sharing